

Test Case ID	Target Feature	Outcome Before Refactoring	Outcome After Refactoring	Engineering Performance Improvement / Observation
TC-01	Basic Element Insert	Functional but used global variables.	Functional with encapsulated class.	Data is safely managed inside the <code>DynamicArray</code> class.
TC-02	Dynamic Memory Resize	Memory leak occurred during resizing.	Old memory is released using <code>delete[]</code> .	Prevents heap memory leaks during capacity expansion.
TC-03	Traversal Output	Off-by-one heap over-read.	Clean traversal from <code>0</code> to <code>size - 1</code> .	Prevents invalid memory access and garbage values.
TC-04	Linear Search	Worked with unnecessary inner loop.	Works using one loop.	Cleaner $O(n)$ search with less unnecessary overhead.
TC-06	Element Deletion	Worked but contained redundant nested loops.	Uses a single-pass $O(n)$ shift.	Simpler and more efficient deletion routine.
TC-07	Negative Index	Negative index was rejected.	Invalid index is safely rejected.	Maintains safe input validation.
TC-08	Upper Boundary Check	Could access invalid memory and crash.	Invalid index is rejected safely.	Prevents out-of-bounds access and possible crashes.
TC-09	Matrix Transpose	Correct for 3×3 matrix.	Correct for 3×3 matrix.	Matrix logic remains $O(R \times C)$ but is given a

				clearer function name.
TC-10	Program Cleanup	Dynamic memory was leaked.	Destructor releases allocated memory.	RAII provides automatic memory cleanup.